

Integrating ethical considerations and community engagement for using genomic data in Africa – KidneyGenAfrica

Day 5 – first session: KidneyGenAfrica: 1st Training
Workshop

Michele Ramsay with assistance from Michelle Bishop

30th January 2026

University of the Witwatersrand, Johannesburg, South Africa

“Too much meat does not
spoil the soup”

Ghanaian proverb

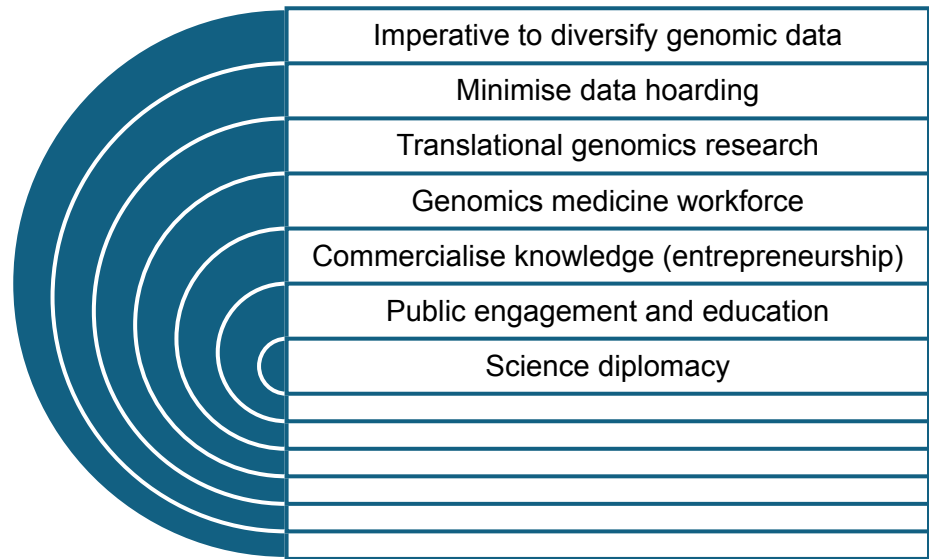
Importance of genomic data sharing for Africa

Current Genetic Medicine Reports (2025) 13:1
<https://doi.org/10.1007/s40142-024-00211-0>

REVIEW

Science and Society: Pathways to Equitable Access and Delivery of Genomics Medicine in Africa

Nchangwi Syntia Munung¹



COMMENT | 08 September 2025

Can a bold ‘social contract’ make data sharing more palatable?

Hoarding of health data is common across Africa, owing to well-founded concerns about exploitation and misuse of biomedical information. A charter that guarantees responsible data stewardship would help to allay such fears.

By [Nchangwi Syntia Munung](#)✉, [Cornelius Ewuoso](#)✉, [Francis E. Agamah](#) & [Emile R. Chimusa](#)✉

Objective and outline

- Objective: To provide information on ethical practices and community engagement for large genomics projects (African contexts) and best practices for data sharing and use
- Outline
 - WHO principles for genomic data collection, use and sharing
 - Building research cohorts
 - Data sharing and Trusted Research Environments (including federated data analysis)
 - Publishing in the field of Precision Medicine – a helpful checklist
 - Navigating access to international databases with genomic data

WHO Principles for genomic data collection, use and sharing

WHO Science Council



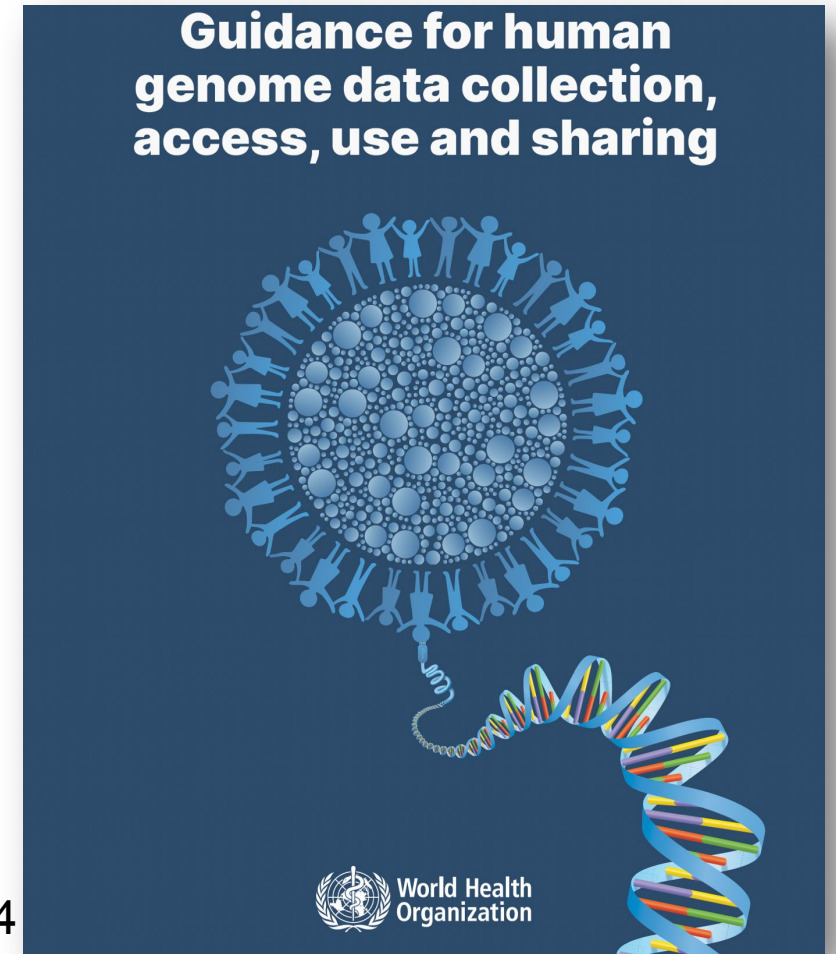
“Current uses and future applications of genomic technologies are critical for improving the health and livelihood of people in all parts of the world, regardless of economic status”



WHO releases new principles for ethical human genomic data collection and sharing

July 2022

November 2024



<https://www.who.int/publications/i/item/9789240052857>

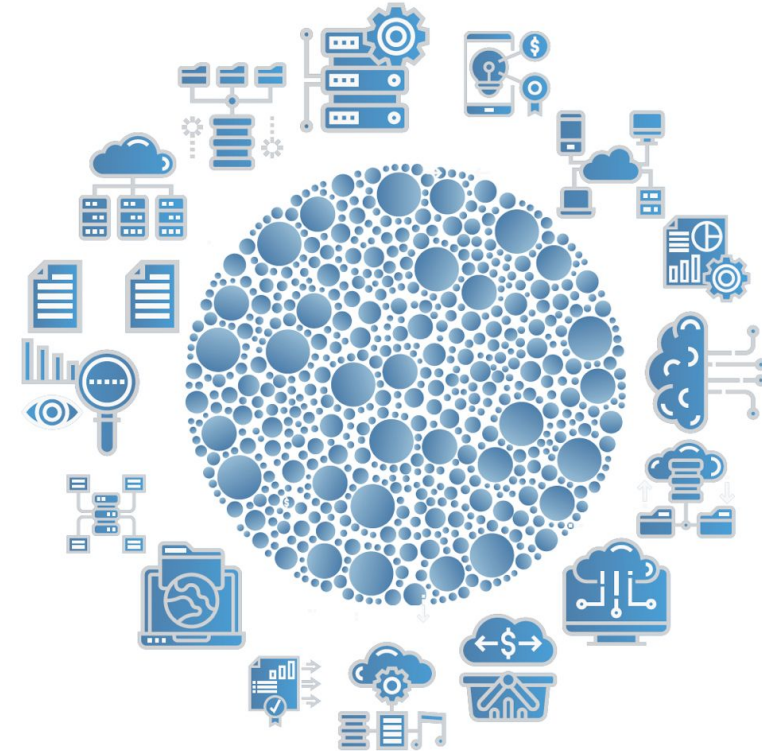
<https://www.who.int/publications/i/item/9789240102149>

If you were developing principles for genomic data collection, access, use and sharing, what would you prioritise?

Give your top 4 priorities

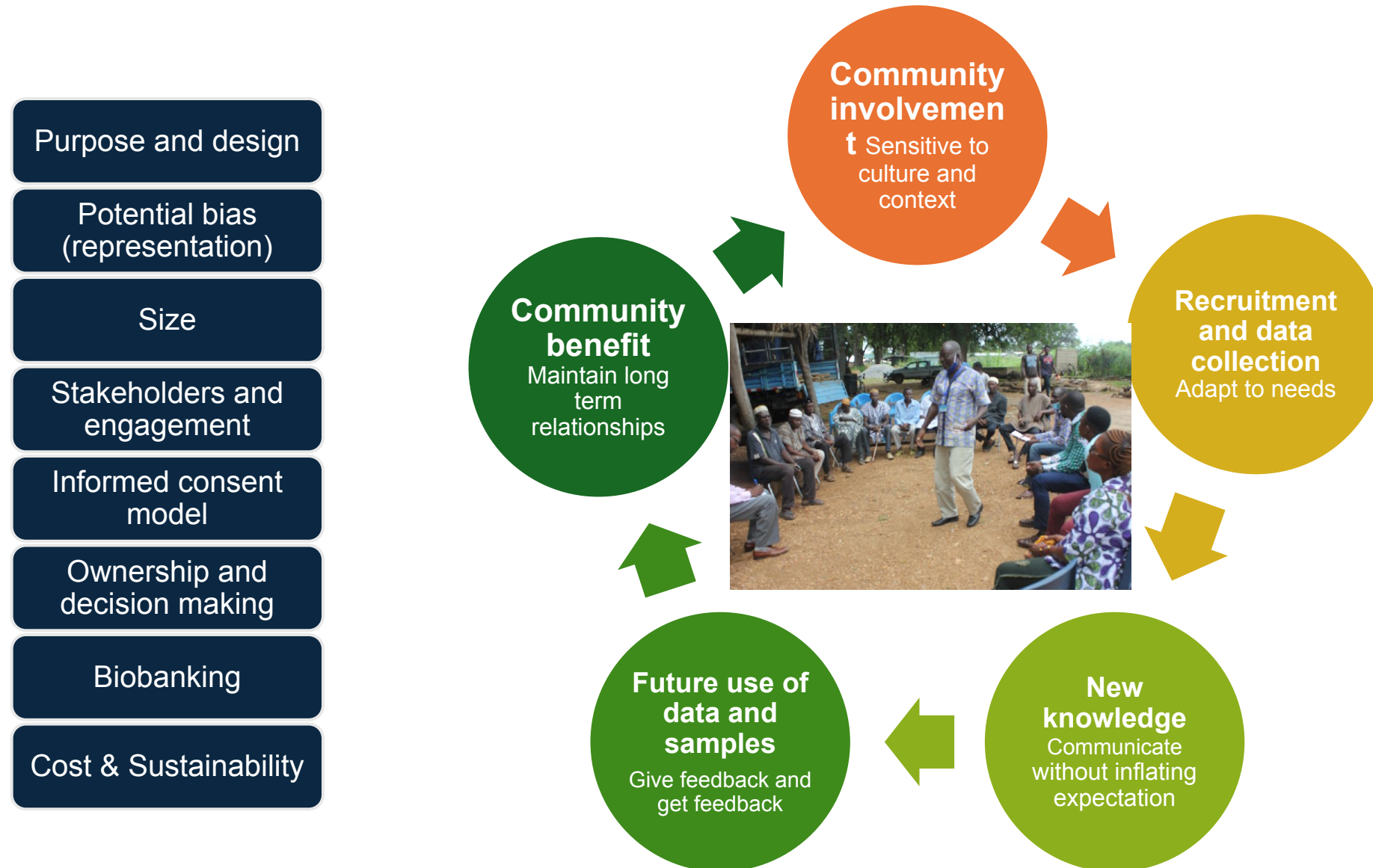
Guiding principles for human genome data collection, access, use and sharing

- ❖ Value rights of individuals and communities to make decisions
- ❖ Social justice
- ❖ Solidarity
- ❖ Equitable access to benefit
- ❖ Collaboration, cooperation and partnership
- ❖ Stewardship of human genome data
- ❖ Transparency
- ❖ Accountability



Building research cohorts

What to consider when planning a cohort



Consent models

Specific consent	Consent for specific research purposes without permissions for secondary use
Broad consent	Consent that allows the use of human biological samples and associated data for future research, supported by a governance framework regulating secondary use
Tiered consent	Presents potential participants with a menu of research categories to which they may consent
Dynamic consent	A personalized, digital communications interface that enables two-way communication between participants and researchers, and puts participants at the centre of decision-making
Blanket consent	Consent for unspecified use of samples and data without restrictions

Participant autonomy:
Informed consent and informed decision making

What reasons may be given to decline participation in a study?

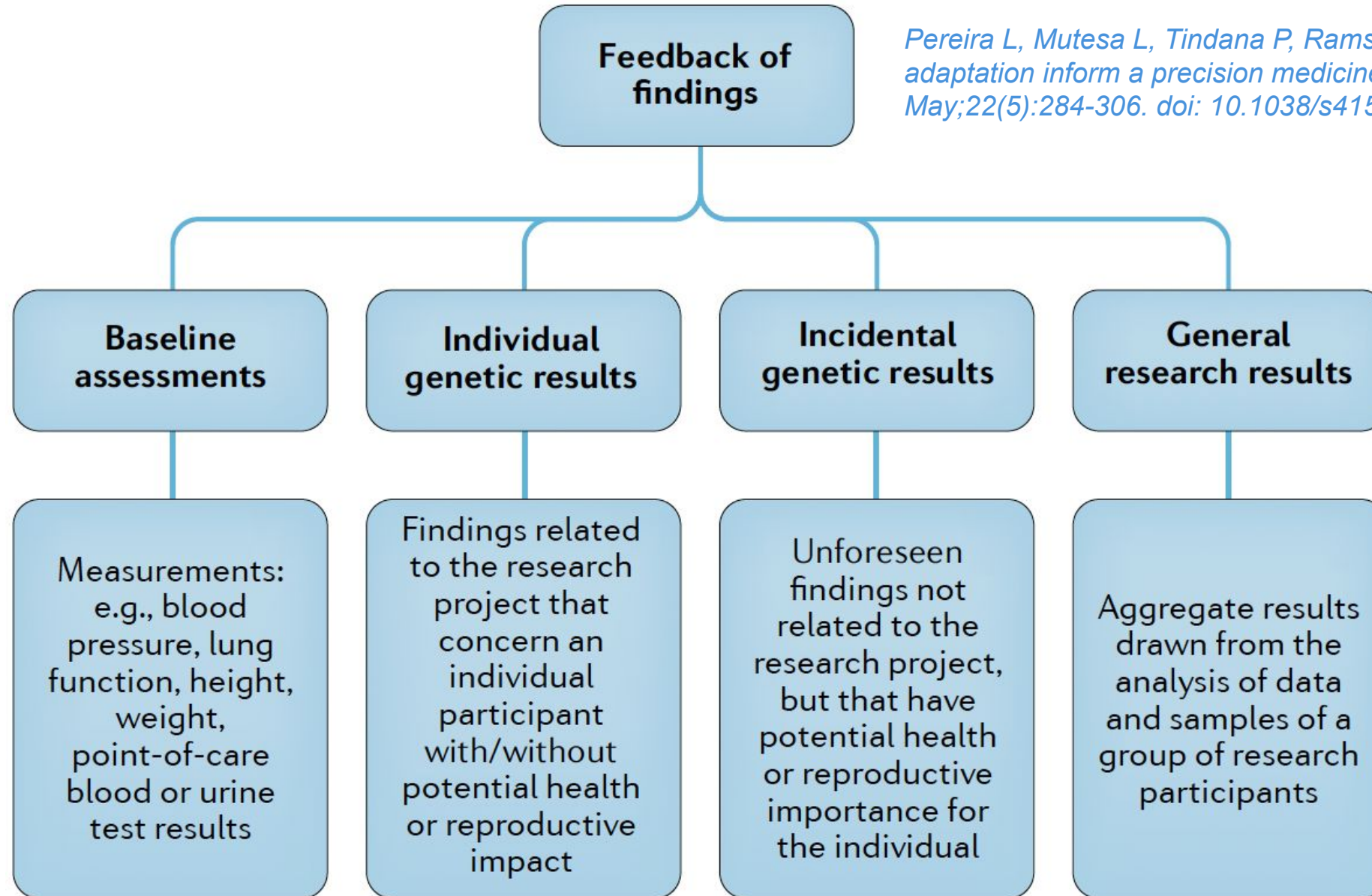
What are the benefits of the
broad consent model?

Give 3

What are the potential pitfalls
of the broad consent model?

Give 3

Pereira L, Mutesa L, Tindana P, Ramsay M. African genetic diversity and adaptation inform a precision medicine agenda. Nat Rev Genet. 2021 May;22(5):284-306. doi: 10.1038/s41576-020-00306-8.



If individual level genetic results are returned, what responsibilities, if any, do researchers have?

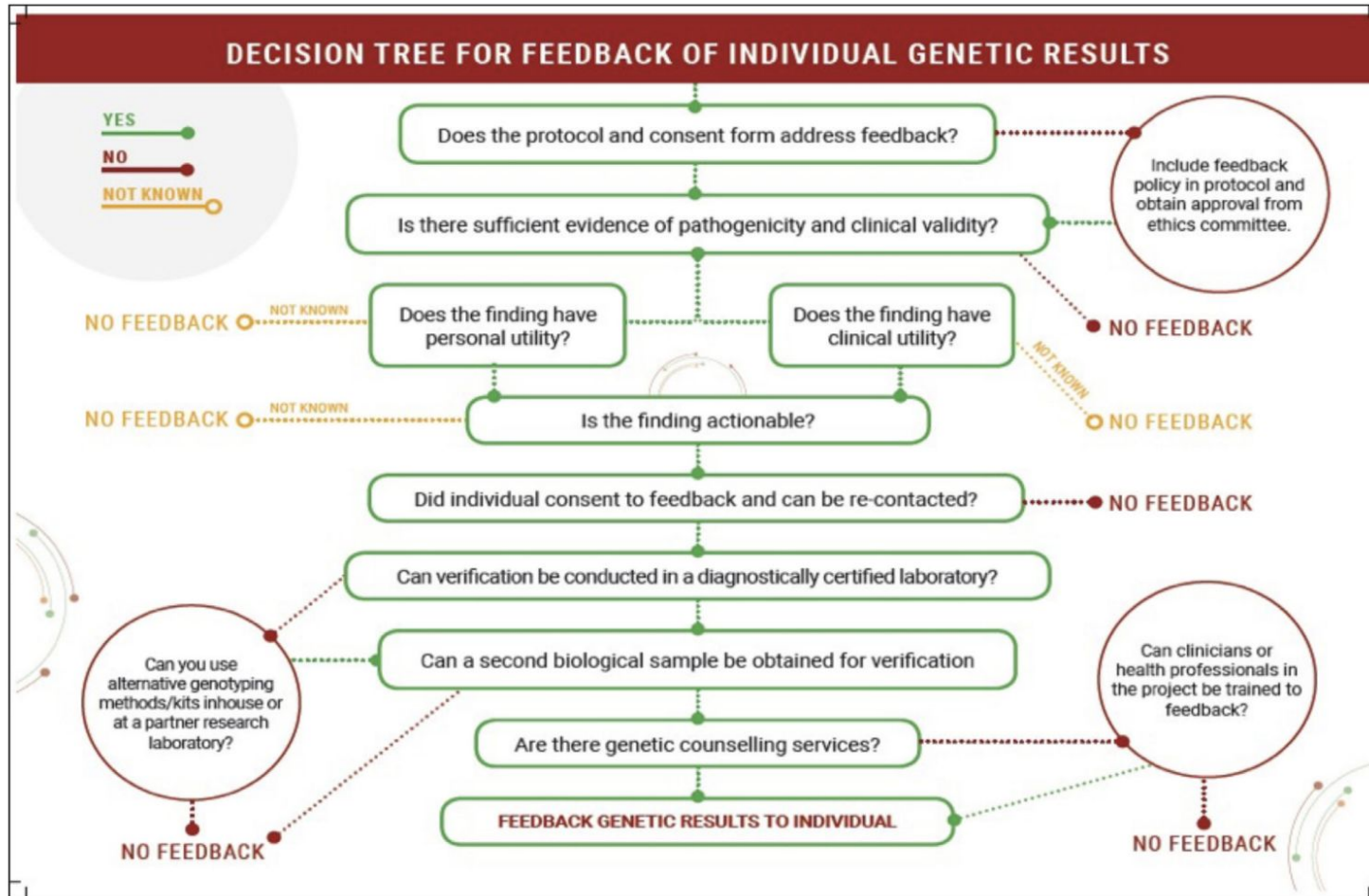
Suggest at least two

For a GWAS study would you
feed back individual level
genetic results?

Yes or No or Maybe

Debate the conditions and considerations

H3Africa – decision tree for individual genetic results



Inform research participants of the risks and benefits

Get ethics approval for the process followed

Ensure that appropriate downstream processes are in place in a clinical setting (genetic counselling, confirmation of result, medical action if required, family cascade screening if considered)

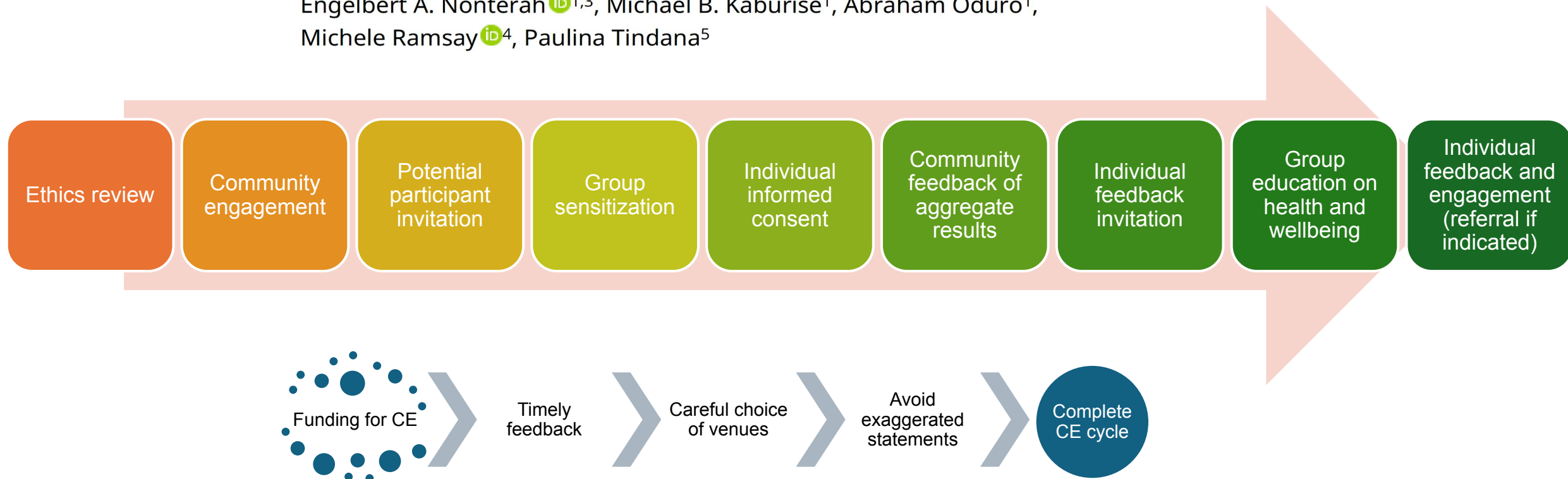
Matimba A, et al. and Community Engagement Working Group. Guideline for feedback of individual genetic research findings for genomics research in Africa. *BMJ Glob Health*. 2022 Jan;7(1):e007184. doi: 10.1136/bmjgh-2021-007184.



OPEN LETTER

Community engagement and feedback of results in the H3Africa AWI-Gen project: Experiences from the Navrongo Demographic and Health Surveillance site in Northern Ghana

Godfred Agongo¹, Cornelius Debpuur¹, Lucas Amenga-Etego²,
Engelbert A. Nonterah ^{1,3}, Michael B. Kaburise¹, Abraham Oduro¹,
Michele Ramsay ⁴, Paulina Tindana⁵



RESEARCH

Open Access



Perspectives on returning individual and aggregate genomic research results to study participants and communities in Kenya: a qualitative study

Isaac Kisiangani^{1*}, Shukri F. Mohamed¹, Catherine Kyobutungi¹, Paulina Tindana², Anita Ghansah³, Michele Ramsay⁴ and Gershim Asiki¹

1. Majority would like to receive feedback on genomic results
2. Researcher obligation to determine how best to do this
3. Communication method appropriate for research results (significance, risk level, and the available types of intervention)

Skillful communication and appropriate language

Participant perspectives on return of results

Recognising participant contribution to research (sign of respect)

Encourage participation in future research

Increased awareness of health status and need to seek medical attention

Increase awareness of scientific progress

How to share - Easily understood language and messages

Timely delivery

Maintain confidentiality when sharing sensitive results

H3Africa – Guideline on Community engagement and consent models

[H3A Biospecimen Sharing Access and Release Policy_1 March 2021](#)

1 April 2020

[H3Africa Recommendations and Guidelines for Genomic Summary Data Sharing](#)

25 June 2020

[H3Africa Report on Informed Consent and Commercialization](#)

11 November 2020

[H3Africa Data and Biospecimen Access Committee Guidelines](#)

April 2020

[H3Africa Consortium Affiliate Membership Policy Guideline](#)

March 2020

[Decision Tree for Feedback of Individual Findings](#)

[Guideline for the Return of Individual Genetics Research Findings](#)

[Protocol for Obtaining Core Phenotypes](#)

[Archive Submission Statement](#)

[H3Africa DBAC Managing Conflict of Interest Policy and Guidance](#)

[H3Africa Biospecimen Release MTA Draft](#)

August 2018

[H3Africa Guidelines for Informed Consent](#)

October 2017

[H3Africa Guidelines for Community Engagement](#)

September 2017

[Framework for African genomics and biobanking](#)

February 2017

[H3Africa Phenotype Harmonization Working Group Recommendations](#)

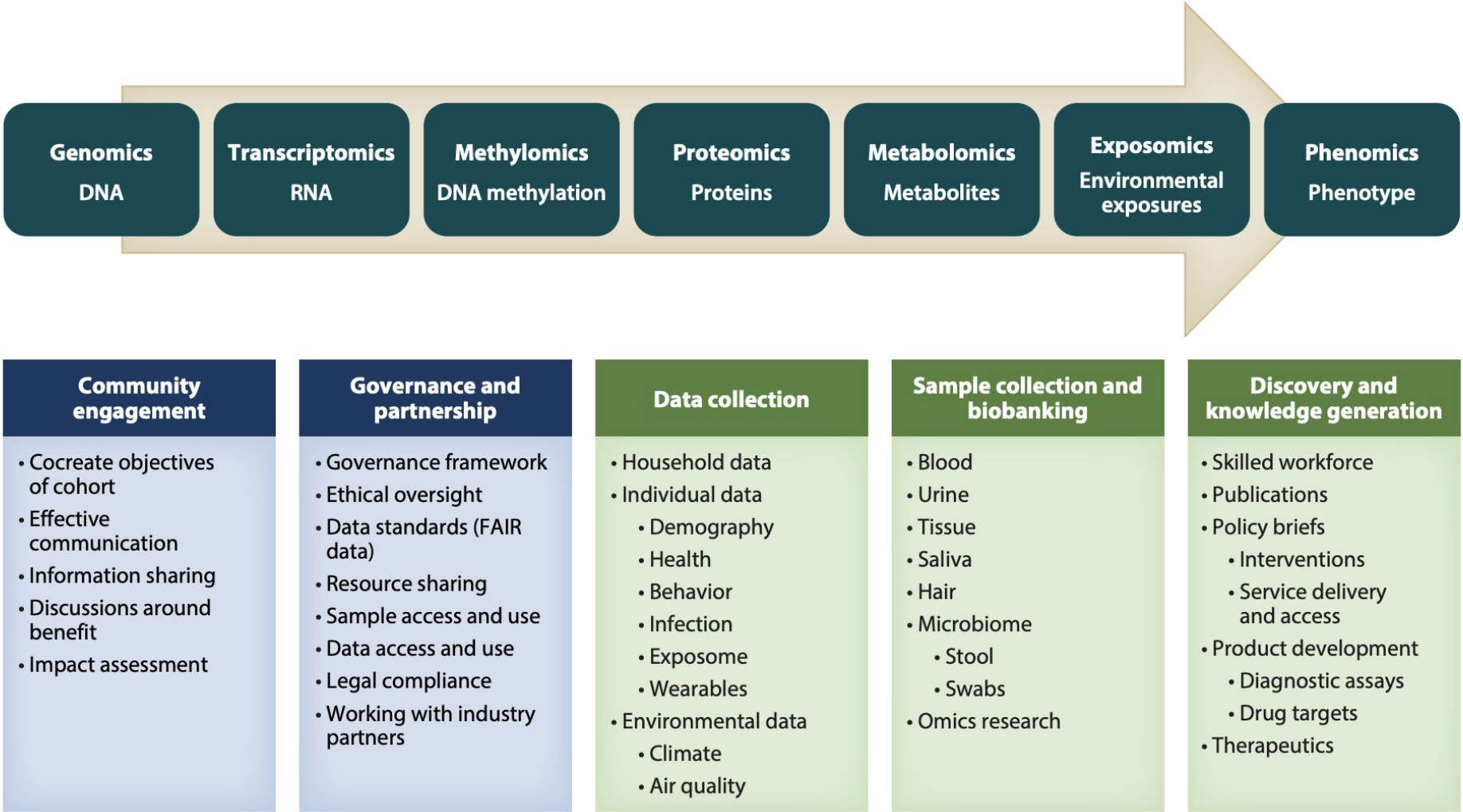
October 2014

[H3Africa Publications Policy](#)

January 2014

<https://h3africa.org/index.php/consortium/consortium-documents/>

Population Cohorts – Multi-omic data – what to consider



The Value Proposition of Coordinated Population Cohorts Across Africa. Annual Review of Biomedical Data Science Volume 7:277-294 (August 2024) <https://doi.org/10.1146/annurev-biodatasci-020722-015026>

National genome projects in Africa

Active National
Genome Projects

- Egypt, Tunisia, Senegal

Government
supported projects in
early stages

- South Africa, Tanzania, Kenya

Emerging National
projects from
Research
Collaborations

- Ghana, Uganda, Botswana,
DRC

Others

- ?

Developing capacity for GWAS cohorts in Africa

Box 3 | Developing capacity for genome-wide association studies in Africa

Building local resources

Obtaining well-characterized phenotypic data from thousands of individuals in poor communities with no systematic medical records requires investment to strengthen infrastructure for clinical research and data management. There also needs to be a commitment to build the resources for genetic and genomic research in Africa, and to foster a cadre of African scientists with the expertise needed to lead this research area. The [African Society of Human Genetics](#) provides an important forum for knowledge-sharing, networking and interactive training programs, with an annual conference held in a different African region each year^{107,108}. The Society also recognizes the need to communicate with policymakers and to attract global attention to the efforts of African scientists, which will be crucial in persuading African institutions and governments to engage with large-scale projects in human genetics and genomics. Another example of capacity building is the [Malaria Genomic Epidemiology Network](#) (MalariaGEN) scheme of data fellowships and data bursaries, which provides training and support in statistics and informatics for researchers in 15 malaria-endemic countries³⁷. Institutions also need to develop infrastructure to manage genetic samples and data within an appropriate regulatory and ethical framework, an example being the Gambian National DNA Bank¹⁰⁹.

Data-sharing networks

Data-sharing networks, such as the [Wellcome Trust Case Control Consortium](#) and the [Genetic Association Information Network](#), have been instrumental in driving forward genome-wide association (GWA) research^{58,110}. Bringing together African and non-African partners will require work at many different levels — for example, reaching decisions about the ownership and permitted uses of shared data and samples, intellectual property and data release, and developing web information systems to integrate data from different research groups and to make these data widely accessible to partners in Africa^{111,112}. The MalariaGEN network has developed policies for data sharing that take account of the disparities of research in high- and low-income countries³⁷. For example, MalariaGEN's policy for GWA data release promotes data access while guarding against uses that might lead to ethnic stigmatization, in addition to setting timelines for data release that are fair to contributing investigators in malaria-endemic countries, who might not be in a position to analyse their data as rapidly as researchers in high-income countries^{37,113}.

Ethical issues

Genetic research in developing countries raises a wide range of ethical and social issues¹¹⁴, and considering them all is beyond the scope of this Review. An example is the problem of obtaining valid consent from study participants for whom terms such as 'genetics' and 'database' might be meaningless unless carefully explained, drawing from local experience^{115–117}. In this situation, it is important to take account of local language and cultural practices and it may be appropriate to seek consent from community leaders as well as from individuals. Another sensitive issue is the fear that the use of information about ethnicity in genetic studies might lead to ethnic stigmatization¹¹⁸.

Towards an African Genome Project

Plans to study African genome variation within the [1000 Genomes Project](#) are currently limited to populations that speak Niger-Kordofanian languages, and will therefore encompass only a fraction of African genetic diversity¹⁰. The African Society of Human Genetics has put forward a strong case for an African Genome Project¹⁰⁸, one component of which would involve systematic sampling of at least 100 ethnic groups across the continent, including minority populations that will be poorly represented in the 1000 Genomes Project. The fundamental aim is to develop population-based resources to investigate genetic and environmental determinants of disease. As discussed above, there is also a need for investments in training and infrastructure. Although these plans remain embryonic, they demonstrate that the time is ripe for an expansion of genome research in Africa, led by African investigators.

Building local resources

- Large well phenotyped cohorts
- Skills development
- Policy engagement
- Infrastructure (biobanks)

Data-sharing networks

- Partnerships
- Policies for data sharing
- Fair opportunity to use data

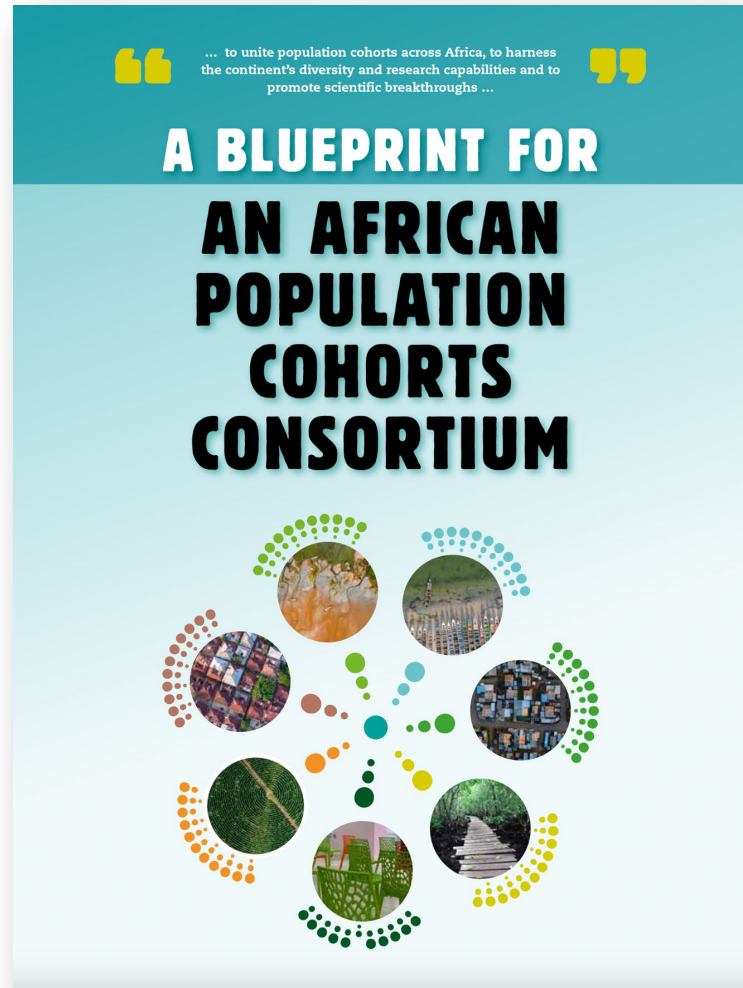
Ethical issues

- Valid consent (language and words)
- Local languages
- Sensitive to cultural practices
- Ethnic stigmatisation

Towards and African Genome Project

- HapMap and KGP only
Niger-Kordofanian language groups
- African Society of Human Genetics

African Populations Cohorts Consortium



<https://ce-apcc.org/wp-content/uploads/2024/05/APCC-B-lueprint-Document.pdf>

Media Centre

£10 million investment in health research towards empowering African genomic discovery

Monday, November 24, 2025

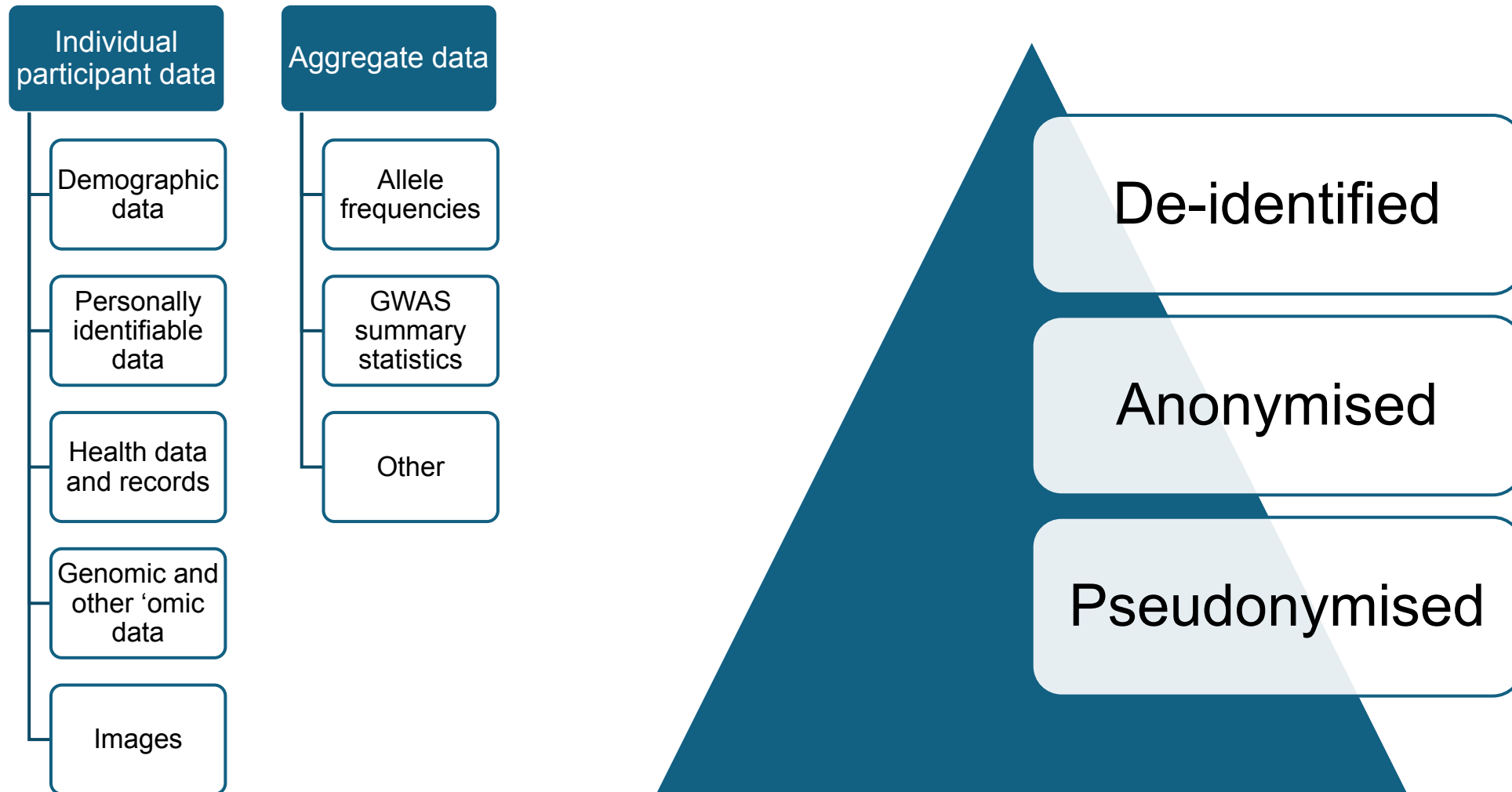
Genomics-led Impact on African Population Health (GEN-IMPACT), a pioneering, Africa-led initiative designed to advance health through genomic discovery

Partnership: The Science for Africa Foundation (SFA Foundation), the African Population Cohorts Consortium (APCC), and the African Bioinformatics Institute (ABI)

To catalyse a 5-year proof-of-concept platform to demonstrate the value of African genomics for health

Data sharing and Trusted Research Environments (TRE)

Levels of data sharing



Which type of data cannot be re-identified?

De-identified, Anonymised or Pseudonymised

Data protection laws in Africa

Munung NS, Staunton C, Mazibuko O, Wall P, Wonkam A. Data protection legislation in Africa and pathways for enhancing compliance in big data health research. Health Res Policy Syst. 2024 Oct 15;22(1):145. PMID: PMC11479556.

Data sovereignty
What does this mean?

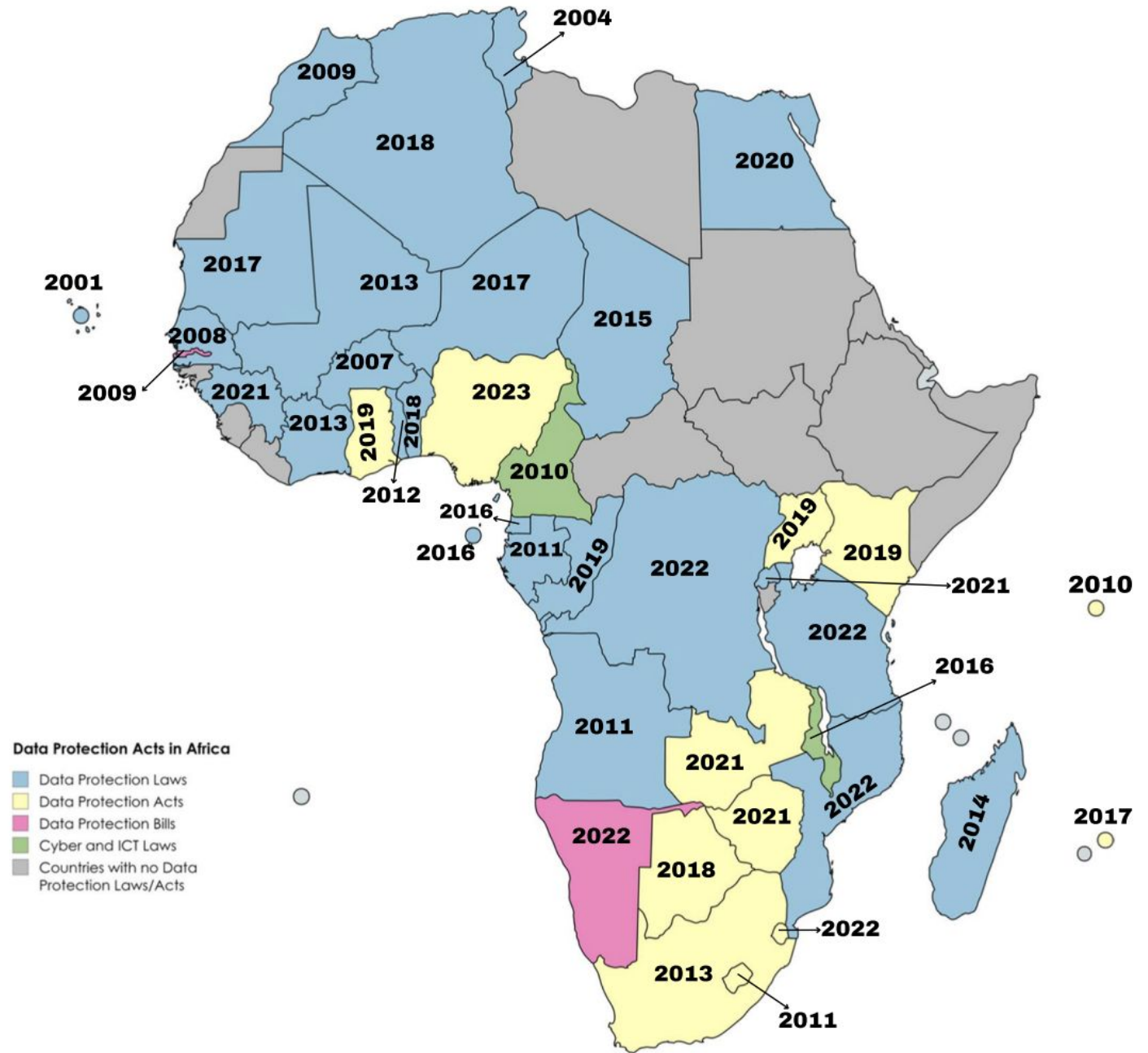
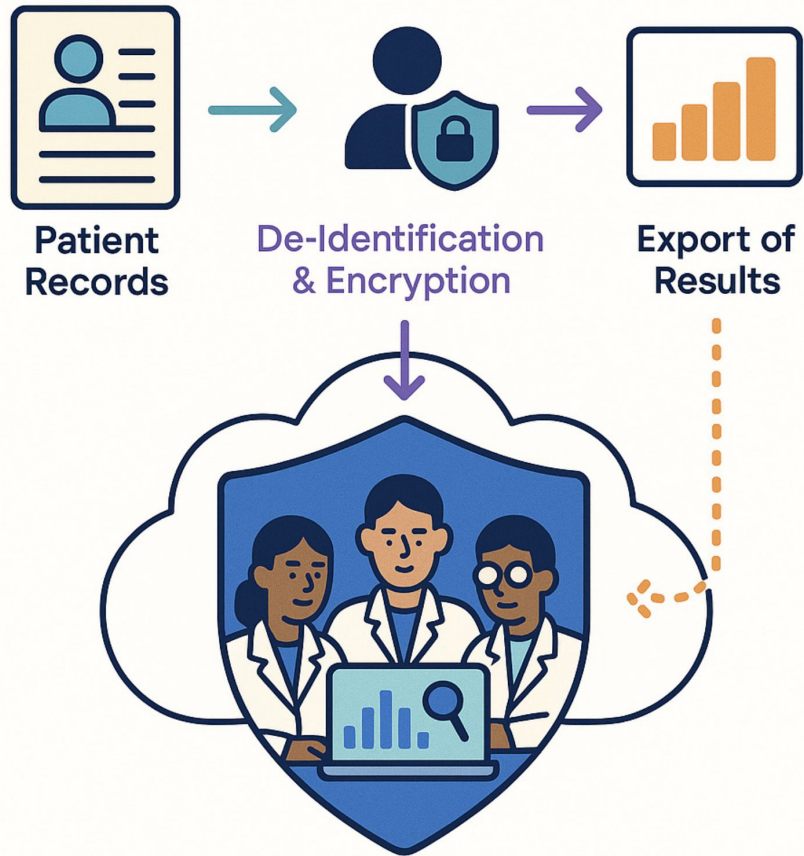


Fig. 1 Representation of African countries with data protection legislation/statutes and year enacted or drafted

What is a Trusted Research Environment?



Highly secure computing environment to provide remote access to (health) data for approved research

Federated data analysis - Play GA4GH viden

**Enabling federated discovery, access, and analysis of
global datasets**



[https://www.ga4gh.org/video/enabling-federated-discovery
-access-and-analysis-of-global-datasets/](https://www.ga4gh.org/video/enabling-federated-discovery-access-and-analysis-of-global-datasets/)

Publishing in the field of Precision Medicine

BePRECISE – Better Precision-data Reporting of Evidence from Clinical Intervention Studies & Epidemiology (2024)

nature medicine

Consensus Statement

<https://doi.org/10.1038/s41591-024-03033-3>

Reporting guidelines for precision medicine research of clinical relevance: the BePRECISE checklist

Why?

- ❖ No standardized reporting in PM publications
- ❖ Improve accuracy and reduce errors in medical and health recommendations, while maintaining safety and cost-effectiveness

Item number	Item wording	Elaboration and explanation of item
E. Equity, inclusion, diversity and PPIE. Authors are encouraged to address these topics in their manuscripts within relevant sections. The reporting items listed herein are not exhaustive and all considerations of PPIE (including patient-reported outcomes and experiences), as well as any community engagement efforts, should be described wherever possible.		
E1	Use appropriate population descriptors such as ancestry, geographic and sociodemographic characteristics of all participants, particularly those in under-represented groups	In cases where data from under-represented group(s) are collected and the subgroup size is $n \leq 20$, all data should be analyzed and reported (even in cases where subgroup analyses might be considered underpowered, because this will facilitate subsequent meta-analyses of results). A minimum sample size of 20 is based on the All of Us Research Program Data User Code of Conduct (https://www.researchallotment.org/precision-user-code-of-conduct/) and is intended to avoid disclosing individual participant identity. Avoid merging subgroups into larger heterogeneous groups (for example, 'non-European ancestry'). Although there is ongoing discussion on the appropriate use of words and terms describing groups within populations, this checklist yields to other guidelines on this matter. If data pertaining to race and/or ethnicity are collected this should be reported in accordance with relevant established guidance.
E2	Describe the implications of inclusion and/or exclusion of people who are under-represented in precision medicine research or underserved by health services	Describe implications for successful extrapolation of study findings to other groups, particularly those typically under-represented in precision medicine research
E3	Describe PPIE in any aspect of the study design, conduct and/or reporting	PPIE may include consultation, involvement, partnership or leadership by end-users, including being part of the research and/or authorship team
E4	Where possible, and ideally with guidance from PPIE research services, describe the potential impact of the study's results from a lived-experience perspective, especially the impact of the research on people living with disease	
1. Title and/or abstract		
1.1	Include 'precision medicine' in the title or abstract	These reporting guidelines use the terms 'precision medicine' and 'personalized medicine', defined elsewhere ¹ , as follows: Precision medicine focuses on minimizing errors and improving accuracy in medical decisions and health recommendations. It seeks to maximize efficacy, cost-effectiveness, safety, access for those in need and compliance compared with contemporary medical practice. Precision medicine emphasizes tailoring diagnostic or therapeutic (prevention or treatment) to subgroups of populations sharing similar characteristics. The use of a person's own data to objectively gauge the efficacy, safety, and tolerability of therapeutics, and, subjectively, to tailor health recommendations and/or medical decisions to the individual's preferences, circumstances, and capabilities.
1.2	State the research question and study design	'Study design' refers to the specific type of clinical trial design (for example, parallel arm, randomized crossover, recall-by-genotype) or observational cohort design (for example, cross-sectional study, prospective cohort study, case-cohort study, case-control study), if the study design involves time-series assessments this should also be highlighted
1.3	Describe whether the study relates to prevention, diagnosis, treatment and/or prognosis	
1.4	Describe the population or subgroup that is the focus of the current analyses	
2. Background and objectives		
2.1	State the study hypothesis describing the specific rationale for the precision medicine approach	
2.2	State the study objective(s) of the precision medicine study as (a) etiological, (b) discovery, (c) predictive and/or (d) confirmatory. State all that apply. See the explanation and elaborations document for detailed descriptions of the objectives	(a) Etiological: characterization of heterogeneity across individual level data (b) Discovery: exploration of associations between a set of clinical features and outcome heterogeneity (for example, descriptive RCT subgroup analysis or exploratory analysis of risk factors) (c) Predictive: development of a specific approach(es) to predict heterogeneity in clinical or treatment-related outcomes for individuals or subgroups (d) Confirmatory: reproduction of a previously proposed precision medicine approach
3. Methods		
3.1	Describe aspects of the study design relevant to precision medicine that are necessary for the design to be interpretable and/or replicable	
3.2	Provide the rationale for choice of outcome(s)	
3.3	If the dataset is a subset of a larger study, describe how and why the subset(s) of participants used in the analysis was selected	
3.4	Define any markers used for stratification or prediction of outcomes in individuals or subgroups	'Markers' in this context could include (and not be limited to) biomarkers, molecular markers and clinical characteristics, as well as societal, economic, geographic and cultural factors

Table 1 (continued) | BePRECISE reporting guidelines for precision medicine research

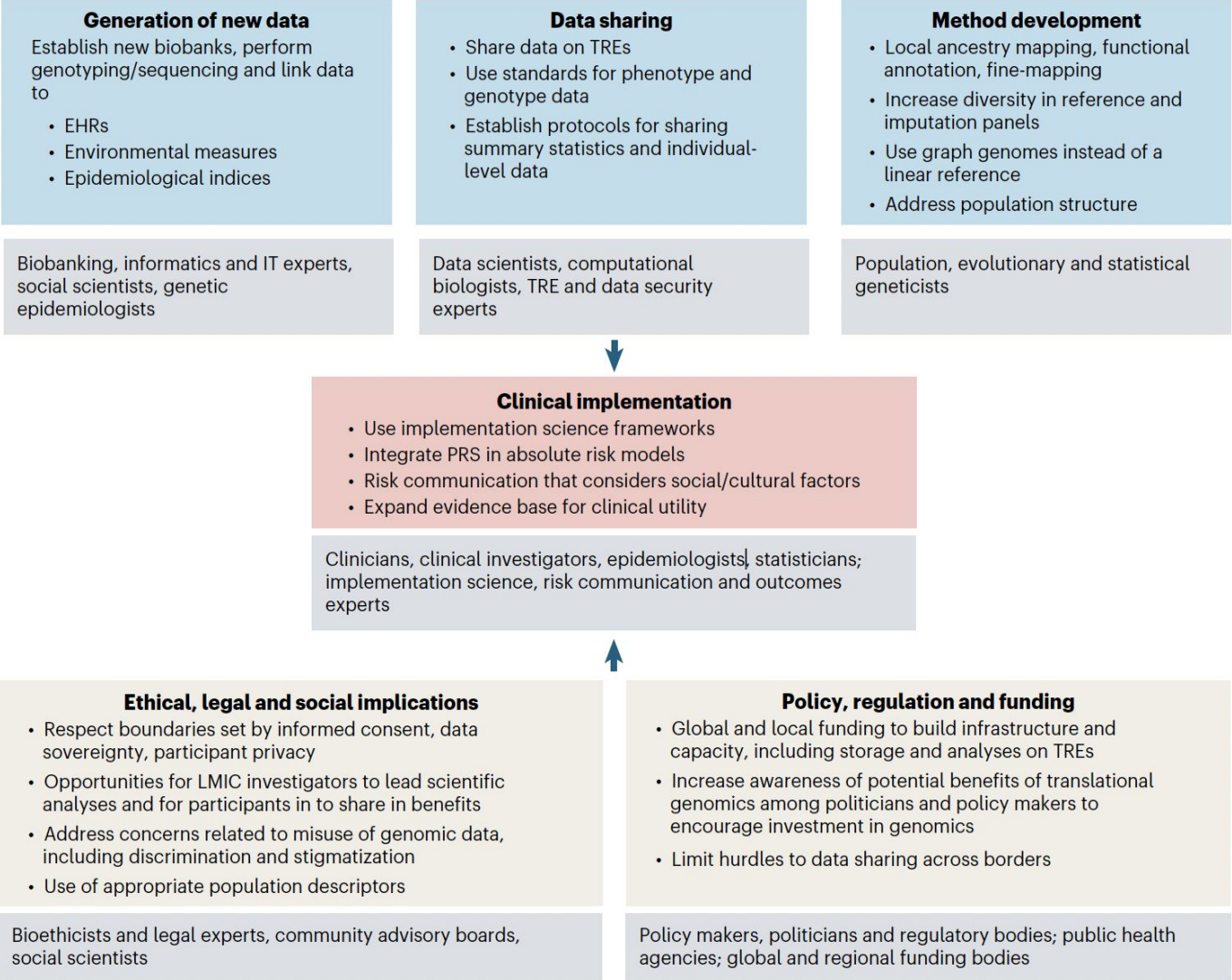
Item number	Item wording	Elaboration and explanation of item
3.5	Provide details of any measures taken to mitigate type 1 and/or type 2 error. Describe a priori power calculations and adjustment for multiple testing, if performed	
3.6	Describe any approach used for internal and/or external replication and/or validation and whether these analyses were planned and relevant datasets identified before or after conclusion of primary analyses	'Replication' analyses are those that seek to directly reproduce primary analyses. 'Validation' analyses are those that seek to generate results using orthogonal methods to those used in the primary analyses that strengthen its conclusions
3.7	Specify how the sample size for any replication/validation study was determined	
4. Results		
4.1	Specify the number of participants in each analysis and provide baseline characteristics	If analysis includes comparison of subgroups, baseline characteristics for each subgroup should be provided
4.2	Report statistical tests and results for subgroup comparisons	Comparisons between subgroups should include appropriate test statistics, which may include tests of interaction and heterogeneity, and in cluster analyses tests of probability for cluster assignment (for example, relative entropy statistic)
4.3	If benchmarking against current practice was undertaken, describe these results. State if benchmarking was not performed	Provide formal comparisons against current practice to assess performance of the precision medicine approach. For example, for prediction models, compare new biomarkers with established prediction variables, formally testing differences in prediction performance. For treatments, compare measures of clinical effectiveness (for example, number needed to treat) between new and conventional approaches. If such comparisons are not possible, provide an explanation
4.4	Provide results for all attempted validation and/or replication analyses	
5. Discussion		
5.1	General limitations	Describe how study characteristics or analytical methods may introduce bias, particularly as these pertain to features of the analysis related to precision medicine (for example, subgroup comparisons)
5.2	Interpretation: describe the precision medicine approach that could potentially be applied in clinical practice	

BePRECISE – Key features

- ❖ Use BePRECISE guidelines with other checklists (CONSORT, STROBE, STORMS)
- ❖ Improve quality and standardization in reporting (accurate comparative data)
- ❖ Accelerate impact of research on target populations
- ❖ Emphasise:
 - Health equity
 - Patient and Public Involvement and Engagement (PPIE)
 - Cost-effectiveness
 - Strengths and limitations
- ❖ Use when planning research, writing manuscripts and reviewing papers

Promoting equity in polygenic risk assessment through global collaboration

<https://doi.org/10.1038/s41588-024-01843-2>



Challenges

Enablers

Implementation

Enablers

Enabling ecosystem

Enablers

Navigating access to international databases

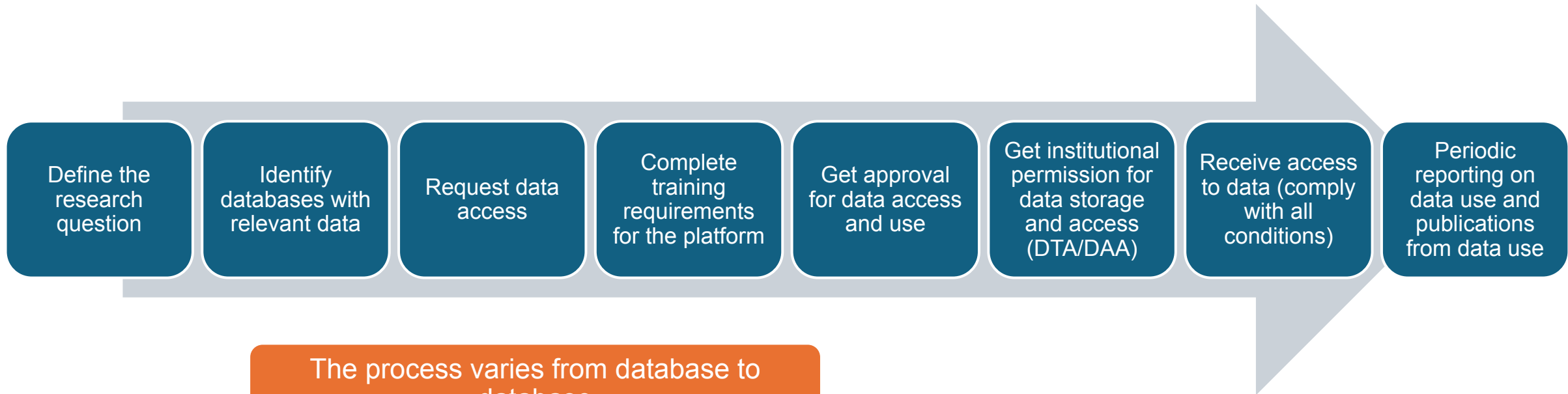
Examples of International Population Cohorts

UK Biobank	• Over 500,000 British participants with whole-genome sequencing (WGS) dataset for research
All of Us Research Program (US)	• Over a million participants, with a strong focus on diversity
BioBank Japan (BBJ)	• Established to research 51 common diseases, includes ~270,000 participants, WGS for a subset
Estonian Biobank	• 200,000 participants (around 20% of the adult population), integrated with the national e-health system
FinnGen (Finland)	• A public-private partnership combining genomic data from up to 500,000 Finns with digital health records to understand disease mechanisms.
PRECISE (Singapore)	• Singapore's National Precision Medicine programme, which aims to sequence 500,000 participants (Chinese, Indian, and Malay ethnic groups)
National Project of Bio-Big Data (NPBBD-Korea)	• Aim to collect clinical and genomic data (WGS) from one million individuals by 2032.
deCODE genetics (Iceland)	• Commercial effort with population data on the Icelandic people, facilitating extensive genetic research
Qatar Biobank (Qatar Precision Health Institute)	• By 2025, ~50,000 participants and WGS released for 25,000

Have you tried to access data
from any of these databases?

Yes or No

Typical workflow to access data



The process varies from database to database

Check eligibility criteria and fine print in terms of requirements

Honour all commitments to data donors and database management

Process for gaining access could take from 2 to 12 months or more

Why is this necessary?

Keystone Symposium Germany 27-30 April 2026

Closing the data gap in Global Genomics, Medicine and Population Health

Closing the Data Gap in Global Genomics, Medicine & Population Health

April 27-30, 2026 • Herrenhausen Palace, Hannover, Germany

KEYSTONE  SYMPOSIA
Accelerating Life Science Discovery

Featured Speakers & Organizers



Lyndon Mitnaul, PhD
Regeneron Pharmaceuticals
Inc.
Tarrytown, NY, USA

Co-Organizer



Naomi Allen, PhD
University of Oxford /
UK Biobank
Oxford, OXF, UK

Co-Organizer



Jeremy Farrar, PhD
World Health Organization
London, UK

Keynote Address



Catherine Kyobutungi, PhD
African Population And
Health Research Center
Nairobi, Kenya

Day 3 Keynote Address



Jessilyn Dunn, PhD
Duke University
Durham, NC, USA



Segun Fatumo, PhD
London School of Hygiene
& Tropical Medicine
London, UK



Melissa A. Haendel, PhD
University of North Carolina
Chapel Hill, NC, USA



Mark I. McCarthy, MD
Genentech, Inc.
South San Francisco, CA, USA



Geoffrey S. Ginsburg, MD, PhD
National Institutes of
Health
Bethesda, MD, USA

Organizer



Michele Ramsay, PhD
University of Witwatersrand
Johannesburg, South Africa

Co-Organizer / Day 2 Keynote Address

#KSPopulationHealth26